Daniele Pessina

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Education

Imperial College London 2022 – Present

PhD in Chemical Engineering - CASE Studentship with AstraZeneca

- Thesis title: A hybrid *in-silico/in-vitro* approach for integrated design and optimisation of large-molecule crystallisation

Imperial College London 2018 – 2022

MEng in Chemical Engineering, 1st Class Honours

Experience

Modeling & Optimization Scientist May 2026 - Present

Deep Origin, Cell Sims team - ARPA-H CATALYST program (PREDICTS consortium)

- Built an end-to-end differentiable PyTorch pipeline that predicts patient-level drug toxicity, chaining mechanistic sub-models (whole-body PBPK, intracellular signalling, and organ-level QSP) through ML translation layers into one trainable system fed by heterogeneous datasets and calibrated against clinical incidence across patient cohorts.
- Made a stack of black-box mechanistic solvers trainable end-to-end: wrapped AMICI/C++ ODE engines, SBML models, a flux-balance MIP optimiser, and an agent-based model as PyTorch autograd components with pluggable gradient backends, so gradients flow from the toxicity prediction back into the neural weights.

PhD Researcher, Process Systems Engineering 2022 - Present

Supervision: Dr. Maria Papathanasiou & Prof. Jerry Y. Y. Heng

- Developed physics-informed neural ODEs with transfer learning to model sparse, multi-fidelity crystallisation time-series, matching from-scratch accuracy with half the training data.
- Applied neuro-symbolic methods to recover interpretable, lower-complexity models of crystallisation dynamics from data.
- Built autodifferentiable probabilistic inference pipelines in JAX and Julia for uncertainty-aware parameter estimation and robust optimisation of PDE- and ML-based process models.
- Built hybrid pharmacokinetic models fusing mechanistic PK equations with neural networks trained on multimodal clinical data (dosing records, EEG signals, patient covariates) for personalised Remifentanil modelling, patient stratification, and flagging anomalous trajectories under distribution shift.
- Designed ML-enhanced global sensitivity analysis and model-reduction tools for intractable crystallisation and PK models.
- Ran *in-vitro* lysozyme crystallisation experiments (Mettler-Toledo EasyMax, PAT tools) that produced evidence for template-induced protein nucleation.

Data Scientist Intern Jan. - Apr. 2026

Quaisr, London

- Partnered with pharmaceutical clients to build and deploy agentic AI workflows for regulated environments, pairing LLM automation with domain data science across the full process-development cycle.

Consultant Data Scientist March - July 2025

Haleon

- Delivered ML soft-sensors that estimate product quality in real time from high-frequency manufacturing data to within 10% of lab reference, enough to displace up to half of offline batch QA checks.
- Compared LSTM and transformer architectures for the task and used data-augmentation strategies to train on heavily imbalanced production data.

Technical Skills

Core: SciML, probabilistic programming, ODE/PDE numerical modelling, Bayesian inference and optimisation, uncertainty quantification, differentiable & multiscale modelling, PBPK/QSP simulation, interpretability (SHAP, PDP/ICE), time-series modelling, surrogate modelling, GPU acceleration

Software: Julia, JAX, Python, PyTorch, AMICI, PEtab, XGBoost, Optuna, MATLAB, R, Git, Databricks, SQL, Docker, Azure

Publications

Pessina, D. *et al.* (2026) Transfer Learning of Data-driven Crystallisation Processes via Constrained Neural Ordinary Differential Equations; *Dig. Chem. Eng.* **(Editors' Choice, March 2026)**

Pessina, D. *et al.* (2026) An in silico/in vitro approach for uncertainty-aware hybrid models for template-induced protein crystallisation systems; *Systems & Control Transactions*

Heng, J. *et al.* (2026) Biomolecular Crystallisation Through Soft Templates and Seeding; *Adv. Biochem. Eng. Biotechnol.* (Springer)

Pessina, D. *et al.* (2025) Model-based approach to template-induced macromolecule crystallisation; *Systems & Control Transactions*

Pessina, D. *et al.* (2025) Machine learning-enhanced Sensitivity Analysis for Complex Pharmaceutical Systems; *Systems & Control Transactions*

Pessina, D. *et al.* (2025) Integrated In Vitro/In Silico Uncertainty Quantification Method for Protein Crystallization Models; *Industrial & Engineering Chemistry Research*

Grob, A. *et al.* (2024) Mammalian cell growth characterisation by a non-invasive plate reader assay; *Nature Communications*

Open-source contributions

gsax | JAX package for global sensitivity analysis of large-scale dynamic models, over 700× faster than SALib on multi-output workloads.